

SimForge: Generalization in Autonomous Driving through City-Scale Digital Twins and Simulation

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Abstract—The critical bottleneck limiting AV generalization lies not in architectural capacity but in the breadth and coherence of training distributions across geographies, conditions, and failure modes [1]. We present SimForge, a scalable data generation engine for safety critical and adversarial driving scenarios, addressing three coupled problems: world coverage, scenario relevance, and controlled diversity. SimScene reconstructs simulation ready digital twins and HD maps from aerial and dashcam imagery, grounded in real world risk through an accident and near miss abstraction loop. SimCloud enables scenario authoring through a road relative representation allowing reinstantiation across digital twins, and expands individual scenarios into dataset families by systematically varying environment conditions. The pipeline is paradigm agnostic: it produces distributional infrastructure to empirically test generalization under modular, end to end, or hybrid architectures.

Index Terms—Autonomous driving, simulation, digital twins, scenario generation, dataset expansion, generalization.

I. INTRODUCTION

Autonomous vehicle development depends fundamentally on simulation. The behaviors that determine whether an AV system is safe, such as multiagent conflicts at unprotected intersections, occluded vulnerable road users stepping into traffic, sudden infrastructure failures, degraded visibility in adverse weather, are precisely the behaviors that are rare, costly, and dangerous to collect at scale on public roads [2], [3]. No fleet, however large, can encounter the full tail of safety critical situations through naturalistic driving alone. Simulation offers the only tractable path to systematically generating, repeating, and varying these scenarios. Yet simulation does not automatically confer generalization. A system trained on a narrow set of hand authored virtual worlds, populated with ad hoc edge cases under fixed appearance conditions, inherits the same distributional blind spots as a log replay pipeline, only in synthetic form [1], [4]. The bottleneck, as recent empirical work confirms, is not model architecture but the breadth and coherence of the driving distributions a model actually sees during training [1]. This observation holds regardless of paradigm: whether the downstream consumer is a modular perception and planning stack or a monolithic end to end controller [5], [6], its generalization ceiling is set by the diversity and relevance of its training data.

Three coupled problems explain why current simulation workflows fail to deliver that diversity. First, producing simulation ready digital twins and HD maps remains

labor intensive and tied to a handful of hand authored environments [4], [7], so training distributions are biased toward the few geographies that happen to have been built, and AV performance degrades sharply when deployed on unseen road geometries [8]. Second, scenario authoring is difficult to scale in a way that respects road topology and the statistical structure of real world incidents [9], [10]; without grounding in observed collision and near miss distributions, synthetic data risks overrepresenting common interactions while missing the rare failure modes that dominate actual crash reports [11], [12]. Third, even when good scenarios exist, converting them into large, structured, multimodal datasets with traceable configuration is typically disjoint from the tools used to author and run simulation [13], [14], making it difficult to audit which regions, failure modes, and operating conditions a model has been exposed to, and making reproducible cross paradigm benchmarking impractical, despite recent efforts to derive reactive closed loop evaluation from real world traces [15]. Existing tools address fragments of this problem: mature simulators like CARLA [4], AirSim [16], and LGSVL [17] provide physics and sensing but rely on hand authored worlds; map prior extensions [18], [19] widen spatial coverage but stop short of full reconstruction; neural scene representations [20]–[22] push fidelity but are not codesigned with OpenDRIVE maps or closed loop execution; and scenario languages [23], [24], fuzzing methods [25], and LLM based generation tools [26] improve authoring but keep scenario logic, map semantics, and batch execution weakly linked. The integration gap, from reconstruction through grounded, portable scenarios to traceable dataset families, remains open.

SimForge is designed to close that gap. The system comprises two tightly coupled subsystems. SimScene reconstructs urban scale digital twins and OpenDRIVE HD maps from aerial and dashcam imagery, augments them with semantic, risk, and visibility metadata, and grounds parameterized scenarios in real world accident and near miss patterns, ensuring that generated situations reflect observed failure distributions rather than arbitrary edge cases. SimCloud authors scenarios through a road relative intermediate representation that decouples scenario logic from world specific coordinates, so a single authored scenario can be reinstantiated across different digital twins without manual rework; it then expands each scenario into dataset families through systematic environmental sweeps, with every run

governed by a configuration manifest that binds scenario, sensors, and conditions to outputs for full reproducibility. A single tile can host dozens of road relative scenarios (e.g. 84 on one deployed Palo Alto asset in Sec. IV-B), each realized under many independent manifests, yielding orders of magnitude more multimodal, traceable realizations than fixed single condition or hand tuned simulation runs. The pipeline is deliberately paradigm agnostic: it produces the distributional infrastructure needed to empirically test whether broader geographic coverage, scenarios grounded in incidents, and systematic condition variation actually close the generalization gap, regardless of whether the model consuming that data is modular, end to end, or hybrid.

II. SIMSCENE

A. Digital Twin Generation

SimScene reconstructs urban scale, simulation ready digital twins from widely available imagery, such as aerial photography (orthographic, nadir, oblique) and dashcam or street level sources, rather than proprietary fleet data [27], [28]. The pipeline proceeds through seven stages: **(i) Multi view photogrammetry** converts registered aerial and dashcam imagery to dense, georeferenced point clouds [29]; **(ii) Geometric cleanup and ground separation** isolates building clusters via sequential multiplane RANSAC and statistical outlier removal; **(iii) LOD2 building reconstruction** recovers watertight CityGML meshes with planar facades, α -shape roofs, and closed ground polygons, from aerial point clouds; **(iv) HD map extraction** derives OpenDRIVE lane networks and GeoJSON features coregistered with the 3D layout [30]; **(v) Facade and prop enrichment** adds doors, windows, and streetscape assets via selected manual refinement; **(vi) Procedural vegetation** populates sidewalks and medians; and **(vii) Asset graph export** packages the scene as OpenUSD/FBX for direct ingestion by engines such as CARLA [4]. The resulting twin is intended to be both *geometrically* and *photometrically* faithful to the imaged site: metric structure and HD maps from photogrammetry and coregistration, with facade, prop, and texture enrichment for plausible closed loop rendering. Maps are further augmented with structured metadata encoding semantic, risk, visibility, and language aligned annotations to support scenario placement and dataset balancing [31].

Stage (iii) is the geometric core. The key challenge is that aerial scanners illuminate rooftops densely but facades sparsely, so naive surface reconstruction fails. Ground separation runs sequential multiplane RANSAC (iteratively fitting and removing planes) and selects the lowest elevation result as terrain, guaranteeing reliable isolation regardless of point density. Footprint extraction projects roof edge points to an axis aligned frame, fills a morphologically closed occupancy grid, and classifies shapes as rectangular (4 vertices) or L-shaped (≥ 6 vertices). Facade panels are synthesized per footprint edge with column heights clamped against a KD tree lookup over roof points to prevent facade and roof penetration; an α -shape roof mesh and fan triangulated

ground polygon complete the watertight building shell. The output is a coherent world package with HD maps, building meshes, props, and simulator ready scene files, enabling new geographic regions to be onboarded without manual rebuilding.

B. Grounding Scenario Generation with Real World Data

Synthetic scenarios decoupled from real failure statistics may overrepresent common interactions while missing the rare failure modes that dominate actual crash reports [11], [12], [25]. SimScene closes this gap through a four layer grounding architecture.

Safety Event Memory. Raw driving events, such as crashes, AV disengagements, and near miss reports [11], [32], are ingested and normalized through a progressive abstraction chain: *RawEvent* $\rightarrow$ *ScenarioPrimitive* $\rightarrow$ *ScenarioTemplate* $\rightarrow$ *MapScenarioIndex*. Each template captures interaction type, actor kinematics, and parameter ranges while discarding location specific coordinates, and is tagged with provenance and hotspot priors derived from the source corpus. This produces a reusable, evidence backed scenario library grounded in observed failure distributions.

Map Intelligence substrate. The HD map is augmented with a canonical scene graph encoding roads, lanes, junctions, crosswalks, curb zones, and drivability signals. Primitive feature detectors populate per region candidate anchors and pooled scenario zones, and third party enrichment (OpenStreetMap, agency feeds) is merged with confidence and freshness metadata. The result is a queryable world model against which scenario templates can be placed.

Scenario Grounding / MapRAG. A retrieval layer bridges user intent and the world model. Given a scenario request, expressed as natural language or a structured incident query, the layer extracts environment, actors, dynamics, and hard constraints, then executes a hybrid search over map zones, candidate anchors, and scenario templates. Deterministic geometric and semantic filters are applied first; a learned reranking step refines the shortlist and generates explainable candidate locations. The output is a grounded *ScenarioSpec* with actors anchored to road IDs, environment constraints resolved, ready for direct emission in SimCloud’s road relative format (Sec. III-A).

Evaluation feedback loop. Retrieved and generated scenarios are evaluated against the Safety Event Memory in a shared scenario signature space: retrieval acceptance, simulation outcomes, and realism scores are compared against real world accident statistics. Reviewed outcomes feed back into template ranking and coverage calibration, closing the loop between synthetic generation and observed failure distributions.

III. SIMCLOUD

SimCloud consumes simulation ready twins and HD maps from SimScene (Sec. II). Fig. 1 sketches the operational pipeline: geospatial imagery supports twin reconstruction; the twin is augmented with metadata derived from incidents and

maps (crash and near miss abstractions, POI alignment) so scenarios can be grounded in real risk structure rather than ad hoc layouts. From that substrate, road relative scenarios are instantiated in simulation (e.g. CARLA) under manifests, then expanded into multimodal dataset families; the two subsections below detail authoring and environmental expansion.

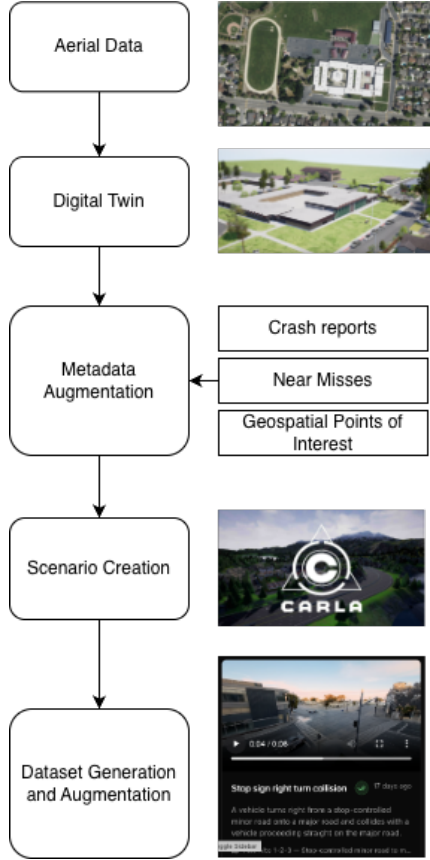


Fig. 1. SimCloud workflow: aerial assets and a digital twin (SimScene); metadata augmentation from crash reports, near misses, and geospatial POIs; scenario creation in simulation; dataset generation and augmentation governed by manifests.

A. Scenario Authoring

SimCloud decouples scenario logic from world specific coordinates through a road relative intermediate representation: each actor is anchored to a road ID, lane ID, and normalized longitudinal position. At simulation time, anchors resolve against the target twin’s OpenDRIVE geometry, enabling a single authored scenario to be reinstantiated across different twins without coordinate rework. Actor behavior is encoded as declarative clip timelines selecting motion primitives with parameters. LLM assisted authoring maps natural language descriptions to actors and behaviors constrained by map metadata [26], or supports interactive refinement through tool calls against road relative state. Each run is governed by a structured configuration manifest binding scenario, sensors, and environment conditions to outputs, ensuring reproducibility and coverage auditability.

B. Dataset Expansion

SimCloud expands each base scenario into a *family* of dataset realizations by sweeping exogenous factors encoded in each manifest, such as illumination and sun angle, atmospheric and precipitation state, seasonal material appearance, traffic and pedestrian density, and sensor pose, while freezing logical actors, road anchors, and motion programs. Parameters are assigned discrete levels or bounded intervals so that factorial or stratified designs yield reproducible coverage over the environmental axis of the generalization problem rather than ad hoc screenshots. Each realization is a separate job with its own manifest, emitting aligned multimodal outputs (RGB, depth, LiDAR, segmentation, trajectories, event logs) for traceable auditing and downstream training under declared distributional assumptions. Engine specific presets (e.g. CARLA weather and sun/sky bundles) and paired transfer or relighting stages (e.g. COSMOS Transfer [33]) are invoked only as backends: the contract to the dataset consumer is the manifest record, not a particular engine string. The generative transfer stage (COSMOS Transfer [33]) applies conditional relighting, material appearance, and weather and surface edits to simulator renders, narrowing the photometric and textural gap between synthetic and fleet captured imagery, a practical sim to real step for perception and prediction models trained on SimForge outputs. Pairing of source and transferred frames under each manifest preserves labels and calibration, so domain shift mitigation remains auditable alongside pure simulation runs.

IV. RESULTS

A. Digital Twin Generation

Fig. 2 summarizes the SimScene LOD2 path in Sec. II-A. (a) Dense reconstruction from registered aerial and dashcam imagery yields a textured mesh over the site. (b) Geometric cleanup and ground separation suppress terrain and clutter so building clusters are isolated for reconstruction. (c) A single building aerial point cloud feeds the LOD2 routine (photogrammetry or fused LiDAR). (d) Footprint extraction recovers the base polygon and eave ring (the seam where vertical facades meet the roof) used to anchor panels in (e). (e) Facade calibration estimates outward facing panels and normals prior to meshing. (f) The classified point cloud separates roof, facade, and eave adjacent samples, reducing roof and facade penetration before (g) merging roof, facade, and ground into a watertight manifold mesh. The automatic LOD2 shell is followed by *selected* manual refinement: artists add higher LOD openings (doors, windows), props, and texture where simulation fidelity requires it.

Scaling to city wide reconstructions demands lightweight meshes. Dense surface methods such as Poisson reconstruction [34] typically yield on the order of 8–10 MB per building, prohibitive when a map tile aggregates hundreds of structures. The LOD2 pipeline (Sec. II-A) instead emits a parametric shell: planar facades, a piecewise roof, and a closed ground polygon.

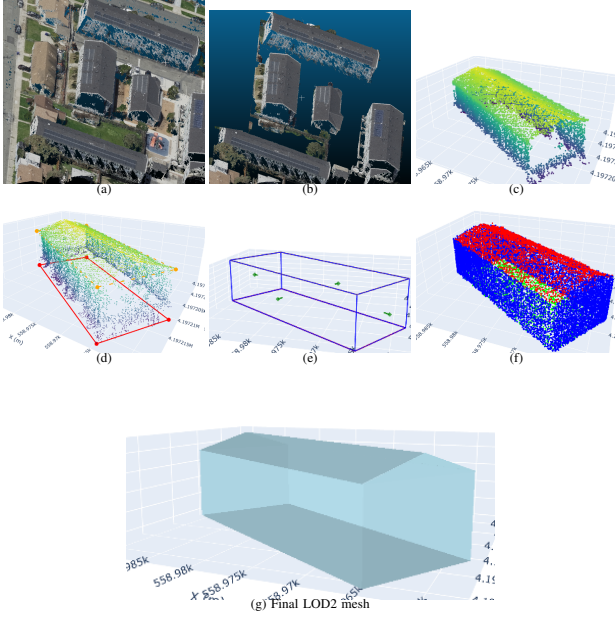


Fig. 2. SimScene digital twin generation (Sec. II-A). From site level photogrammetric mesh and denoised building isolation through footprint/eaves detection, facade calibration, labeled points, to a watertight LOD2 shell; optional manual pass adds fine geometry and textures.

Table I summarizes representative mesh file sizes. Simple rectangular footprints (4 vertices) compress to roughly 1.2 KB, while L-shaped or otherwise complex footprints (≥ 6 vertices) remain in the 2–3 KB range, roughly three to four orders of magnitude smaller, keeping scene graphs tractable at scale.

TABLE I
LOD2 MESH SIZES VS. DENSE POISSON RECONSTRUCTION (PER BUILDING).

Structure type	Poisson	LOD2 (ours)
Rectangular (4 vertices)	8–10 MB	~1.2 KB
L-shape / complex (≥ 6 vertices)	8–23 MB	2–3 KB

Validated against terrestrial LiDAR over a Palo Alto road corridor, SimScene HD maps achieve ~ 10 cm longitudinal MAE and sub 4 cm transverse MAE, sufficient for lane level scenario placement without survey grade infrastructure [27]. At that accuracy, scenarios anchored to lanes and junctions inherit real world spatial constraints at intersection scale rather than idealized placeholders. Representative twins have been imported into CARLA: we execute closed loop runs with spawned actors and multimodal sensors (e.g. RGB and depth), confirming simulation readiness beyond offline mesh inspection.

B. Grounding Scenario Generation

Fig. 3 shows maps augmented with metadata used for grounding (Sec. II-B). (a) On the Palo Alto El Camino Real twin, ten scenario tags, such as unprotected left turns, high rear end risk queue zones, bike and vehicle mixing, index conflict prone structure for map metadata alignment. (b) The map asset library indexes six deployed tiles; one tile (Easterbrook

Discovery School) links 84 authored scenarios and eight tags to HD map geometry, illustrating how families grounded in incidents attach to candidate regions.

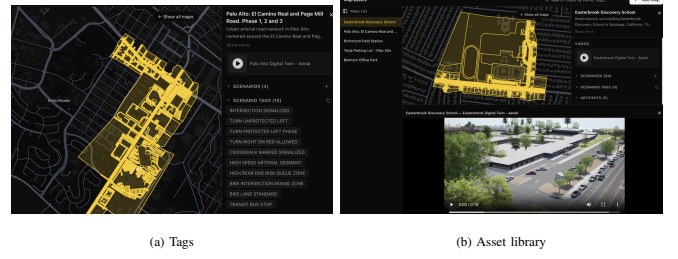


Fig. 3. Grounding: semantic tags on a twin (a) and per tile scenario/tag counts (b).

C. Scenario Authoring

Fig. 4 shows the road relative authoring interface (Sec. III-A). The editor places ego, traffic, and props on OpenDRIVE lanes with Manual or LLM mode; actors are anchored to road IDs before export. For each run, config generation, simulation outputs, and COSMOS Transfer [33] emit artifacts under the same manifest, giving end to end traceability (e.g. Palo Alto parking lot scenario, CARLA).

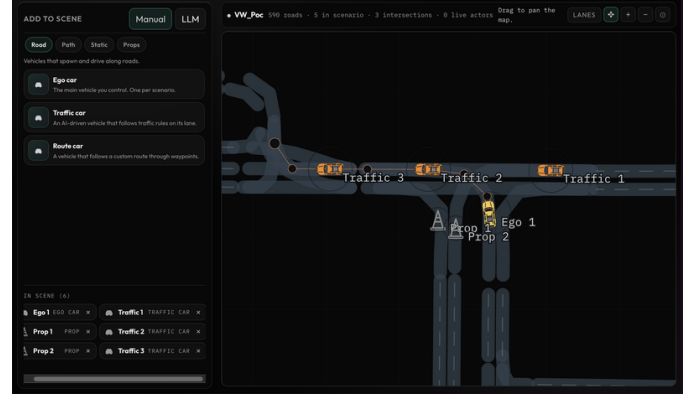


Fig. 4. Scenario authoring: lane anchored editor with actors anchored to road IDs prior to export governed by a manifest.

D. Dataset Expansion

Under sweeps governed by manifests (Sec. III-B), we hold scenario logic and map geometry fixed while systematically varying appearance level factors registered in each run’s manifest: paired illumination and sky state combinations in closed loop simulation (CARLA), and complementary lighting, surface, and weather bundles applied under image space or generative transfer (COSMOS Transfer), in each case with immutable provenance from manifest to outputs. The transfer branch targets sim to real alignment of RGB (and paired modalities) before downstream training or evaluation on real sensors. Fig. 5 shows one authored scenario under multiple ego viewpoints and environmental stresses (e.g. reduced sun elevation, snow covered ground), expanding sensing diversity without respecifying actors or topology.

Fig. 6 complements appearance sweeps with *interaction centric* stress tests that target distinct failure families along the safety critical axis. Panel (a) depicts a low light forward collision regime with constrained photometric margin; (b) a bicycle and motor vehicle conflict amid clutter and lateral uncertainty; (c) a vulnerable road user entering the drivable corridor with partial occlusion; (d) degraded road and tire coupling under standing water and diffuse illumination. Each panel is a qualitative RGB sample from an independent family of runs, each with its own manifest, isolating interaction level adversity orthogonal to pure appearance decorrelation.

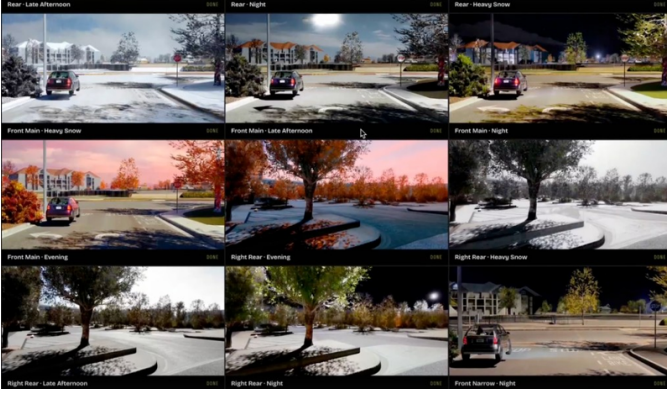


Fig. 5. Dataset expansion: environmental and viewpoint sweeps for one scenario (each panel a separate run governed by its manifest).

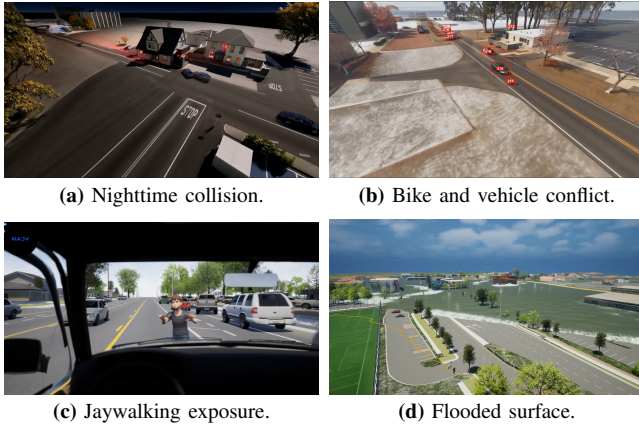


Fig. 6. Dataset expansion: qualitative RGB samples spanning lighting, weather, and interaction adversity (each run under its own manifest).

V. CONCLUSION

Looking ahead, a central goal is to construct a large scale, richly annotated metadata corpus of adversarial driving events spanning the full spectrum of safety critical failure modes: collision kinematics, vulnerable road user exposure, adverse weather and surface interactions, and multiagent conflict at ambiguous right of way boundaries. This corpus is intended to serve as a shared evaluation and training substrate for a broad class of downstream applications, including trajectory prediction under distribution shift, rare scenario fine tuning of deployed AV stacks, safety case validation for regulatory

certification, and closed loop stress testing of modular and end to end controllers alike. By coupling SimForge’s generative pipeline with this adversarial metadata layer, we aim to provide the community with a principled, reproducible path from real world incident evidence to the distributional diversity that safe AV generalization demands. Sweeps governed by manifests are designed to scale data volume superlinearly in the number of environmental and interaction axes: each additional authored scenario or tile compounds into large families of labeled runs, addressing the core “use more data” bottleneck without abandoning traceability. We intend to publicly release selected map tiles, scenario manifests, and expanded dataset families where licensing permits, to support community driven evaluation and cross paradigm benchmarking aligned with open tooling in autonomous driving.

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